

Prim's MST Algorithm:

Prim's algorithm is a greedy algorithm used to find the Minimum Spanning Tree (MST) of a connected, weighted graph.

It starts from one vertex and repeatedly selects the smallest edge that connects a visited vertex to an unvisited vertex.

ALGORITHM:

1. Start with any vertex, usually vertex 0.
2. Mark the starting vertex as visited.
3. Find the minimum-weight edge from a visited vertex to an unvisited vertex.
4. Add this edge to the MST.
5. Mark the new vertex as visited.
6. Repeat steps 3–5 until all vertices are visited.
7. Add all selected edge weights to get the minimum cost.

Time Complexity:

Case	Time Complexity
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Best Case	$O(n^2)$
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Average Case	$O(n^2)$
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Worst Case	$O(n^2)$
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Space Complexity:

Space Complexity = $O(n^2)$

OUTPUT:

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Enter number of vertices: 4
Enter the graph:
0 2 3 0
2 0 1 4
3 1 0 5
0 4 5 0
0 - 1 = 2
1 - 2 = 1
1 - 3 = 4
Minimum cost = 7

...Program finished with exit code 0
Press ENTER to exit console.
```

CONCLUSION:

Prim's algorithm was successfully implemented to find the Minimum Spanning Tree of a weighted graph. It selects the minimum-cost edge at each step and connects all vertices with minimum total cost without forming a cycle.